

FACULTY OF ECONOMIC AND MANAGEMENT SCIENCES

DEPARTMENT OF ECONOMICS

ECONOMICS 114 A1

Date:	8 April 2025
Duration:	2 hours
Total Marks:	60
Examiners:	Dr C. Links (Convenor) Dr K. Malindi Dr M. Nchake Mr J. Pretorius Ms L. Dube Ms R. Petersen Ms J. Kazumunu
Internal Moderator	Dr M van Schoor

PLEASE NOTE: ONLY NON-PROGRAMMABLE CALCULATORS ARE ALLOWED.

ANSWER SECTION A's QUESTIONS ON THE SUPPLIED ANSWER SHEET. INDICATE YOUR CHOICE BY MAKING A CROSS NEXT TO THE NUMBER OF THE STATEMENT.

ALSO ANSWER SECTION B's QUESTIONS ON THE SUPPLIED ANSWER SHEET. INDICATE YOUR CHOICE BY MAKING A CROSS NEXT TO THE NUMBER OF THE STATEMENT.

ANSWER SECTION C's QUESTIONS IN THE ANSWER BOOK PROVIDED. START EVERY QUESTION ON A NEW PAGE. SHOW ALL CALCULATIONS.

Key Abbreviation	Meaning	Key Abbreviation	Meaning
AP	Average product	BL	Budget line
GDP	Gross domestic product	IC	Indifference curve
K	Capital	L	Labour
FF	Feasible frontier	MP	Marginal product
MRS	Marginal rate of substitution	MRT	Marginal rate of transformation
P	Price	PF	Production function
Q	Quantity	TP	Total product
w	Wage	Y	Income

SECTION A

TRUE OR FALSE STATEMENTS

[6 marks: 1 mark for a correct answer and $-\frac{1}{2}$ for an incorrect answer]

1. The first industrial revolution contributed to a widening income gap between nations.
 - a) True
 - b) False

Answer: a

2. The Malthusian model predicts that population increases lead to increases in income, resulting in incomes rising above subsistence levels.
 - a) True
 - b) False

Answer: b

3. The MRT represents the slope of the indifference curve, $(\frac{p_y}{p_x})$ which tells us that rate at which the consumer is able to trade one good (on the horizontal axis) for another good (on the vertical axis).
 - a) True
 - b) False

Answer: b

4. In a standard labour-leisure model where consumption is plotted on the vertical axis and free time on the horizontal axis, if wages increase and leisure (free time) is a normal good, then the substitution effect will always lead to a decrease in free time, while the income effect will always lead to an increase in free time, making the overall effect on labour supply ambiguous.
 - a) True
 - b) False

Answer: a

5. The invisible hand game leads to a suboptimal Nash equilibrium where players could improve their outcomes/payoffs if they choose to cooperate.
 - a) True
 - b) False

Answer: b

6. Social dilemmas are defined as Nash equilibria that are Pareto inefficient.

- a) True
- b) False

Answer: a

SECTION B

MULTIPLE CHOICE QUESTIONS

[24 marks: 2 marks for a correct answer and $-\frac{1}{4}$ for an incorrect answer]

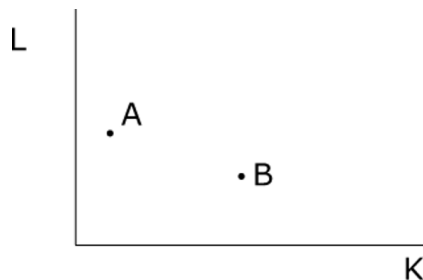
7. Rwanda's nominal GDP in 2024 was 950 billion Rwandan Francs (RWF). Due to the ongoing war with the Democratic Republic of the Congo (DRC), Rwanda's GDP is projected to decline by 45% in 2025. The country's estimated population in 2024 was 102.3 million, but due to mass displacement and emigration caused by the conflict, approximately 50 million people will have left Rwanda by 2025.

Given this information, which ONE of the following statements is INCORRECT?

- a) The population decline between 2024 and 2025 is approximately 48.9%.
- b) The percentage decrease in GDP is smaller than the percentage decrease in population.
- c) GDP per capita in 2025 is expected to be lower than in 2024.
- d) Despite the decline in total GDP, GDP per capita in 2025 will be higher than in 2024.

Answer: c

8. In the graph below, K is the amount of capital required in production and L the amount of labour required in production, and **A** and **B** represent different technologies for producing 20 rugby balls.



A is currently the cheapest technology. However, **B** could become the cheapest technology if

- a) the price of capital and labour both decrease by the same percentage.
- b) no change in prices occurs.
- c) the price of capital and labour both decrease, with a greater percentage decrease for the price of labour.
- d) the price of labour increases while the price of capital remains constant.

Answer: d

9. In 2024, the relative price of a kilogram of maize is 0.5 kilograms of rice. The price of 1 kilogram of maize was 90 Kenyan Shillings (KES).

Which of the following statements is/are CORRECT

- i) The money price of rice is 180 Kenyan Shillings (KES).
- ii) The relative price of rice in terms of maize is 2 kilogrammes of maize.
- iii) The money price of rice is 45 Kenyan Shillings (KES).
- iv) The relative price of rice in terms of maize is 0.5 kilogrammes of maize.

- a) i and ii
- b) iii, iv and ii
- c) i and iv
- d) ii and iii

Answer: a

10. The Allen model of technological innovation argues that Britain escaped the Malthusian trap due to

- a) The cheaper relative cost of labour which created economic incentive for firms to switch to energy intensive technology.
- b) The cheaper relative cost of coal which created economic incentive for firms to switch to energy intensive technology.
- c) Due to better educational outcomes, Britain had a comparative advantage in innovation and saw sustained increases in productivity.
- d) Britain had the cheapest coal.

Answer: b

11. If the MRS of a cup of coffee for bottle of juice is 2, this means:

- a) The consumer prefers coffee to juice.
- b) The consumer is indifferent between a cup of coffee and a bottle of juice.
- c) The consumer is willing to give up 1 bottle juice for $\frac{1}{2}$ cup of coffee.
- d) The consumer would be willing to give up 2 cups of coffee for 1 bottle of juice.

Answer: c

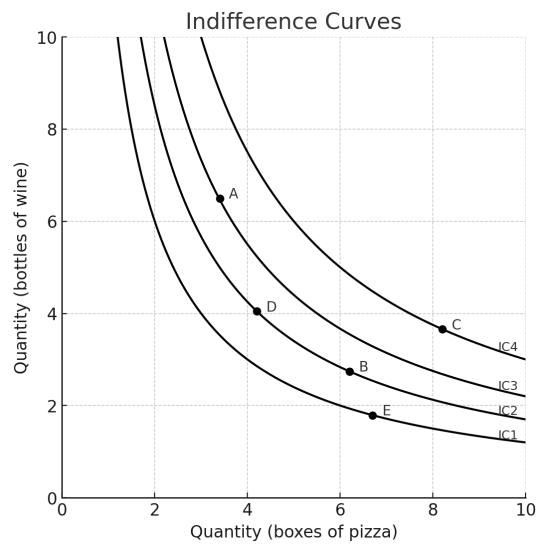
12. David chooses how many hours to spend studying for his A1 test. If the final grade he can achieve is given by a concave production function, this implies that the marginal product of each hour he uses to study _____ as the number of hours of study _____.
- a) decreases, increases
 - b) increases, decreases
 - c) decreases, decreases
 - d) increases, increases

Answer: a

13. A consumer who has a fixed income in Rand, consumes chocolate and noodles. Suppose that chocolate is a normal good, but noodles are an inferior good. When the price of chocolate increases, ceteris paribus, the substitution effect on chocolates is _____, while the income effect on noodles is _____.
- a) positive, negative
 - b) negative, negative
 - c) positive, positive
 - d) negative, positive

Answer: b

14. The figure below represents Sukho's indifference curves for pizza and wine.



Which of the following statements is CORRECT?

- a) Sukho prefers consumption bundle E to B
- b) Sukho is indifferent between consumption bundle A and C
- c) Sukho's most preferred consumption bundle is C.
- d) The marginal rate of substitution of bottles of wine for pizza is higher at bundle B than at D.

Answer: c

Please use the information below to answer Questions 15 and 16

Aqueel and Curtly both like to speed, and like racing on their time off. When they both speed, they have the potential of causing a motor accident and injuring innocent bystanders. When only one speeds, he will win the race and the one that doesn't, will get caught by the traffic officer and be penalized.

		Player 2	
		Don't Speed	Speed
Player 1	Don't Speed	4, 4	1, 5
	Speed	5, 1	2, 2

15. The Nash Equilibrium of the game shown above is:

- a) Don't speed, Don't speed
- b) Don't speed, Speed
- c) Speed, Don't speed
- d) Speed, Speed

Answer: d

16. This game is an example of what type of game?

- a) ultimatum game
- b) invisible hand game
- c) prisoner's dilemma
- d) chicken game

Answer: c

17. In the economy of Namaland, there are two people, Elton and Merle, and two goods, copper and diamonds. Elton likes copper and dislikes diamonds, while Merle likes diamonds and dislikes copper. Due to a halt in mining activities, no copper is available this year, and only 12 diamonds are available due to minimal mining. What is the only Pareto efficient allocation in this economy?

- a) Elton receives 0 diamonds, and Merle receives all 12 diamonds.
- b) Elton receives 6 diamonds, and Merle receives 6 diamonds.
- c) Elton receives 12 diamonds, and Merle receives 0 diamonds.
- d) Any allocation is Pareto efficient.

Answer: a

18. Suppose Warren is coerced into working for Ghislaine on her farm, producing white wine. Warren must give Ghislaine all the extra wine he produces beyond the amount needed for his survival. His production can be represented by a concave feasible frontier, and his needs by a convex biological survival constraint. Which ONE of the following statements is CORRECT?
- a) If $MRT > MRS$, Ghislaine will choose Warren's working hours to be higher.
 - b) If $MRT < MRS$, Ghislaine will choose a lower number of working hours for Warren.
 - c) There will be only one Pareto efficient allocation of surplus between Warren and Ghislaine.
 - d) A Pareto efficient allocation of white wine between Warren and Ghislaine does not necessarily mean that $MRT = MRS$.

Answer: This Question must be dropped